University of Florida, Department of Mathematics

Algebra First-Year Exam, May 2025, Part 1

Answer 4 questions. Indicate which 4 problems you wish to be graded. Write your solutions clearly in complete English sentences. You may quote theorems (within reason) as long as you state them clearly.

1. Let p be a prime and G be a nontrivial finite p-group.

(a) Prove that $Z(G)$ is not trivial.

(b) Prove that every nontrivial finite p-group has a normal subgroup of index p.

2. Let $\sigma \in S_n$ be an n-cycle.

(a) Prove that $\sigma \in A_n$ if and only if n is odd.

(b) Prove that $C_{S_n}(\sigma) = \langle \sigma \rangle$.

(c) Prove that when n is odd, A_n has two conjugacy classes of n-cycles.

3. Let G be a group and let $T(G) = \{x \in G : \exists n \geq 1, x^n = 1\}$.

(a) Prove that if G is abelian then $T(G)$ is a subgroup of G.

(b) Give an example of a nonabelian group G such that $T(G)$ is not a subgroup of G. Justify any claims that you make.

4. This problem has two parts.

(a) Find the order of the group $\mathrm{GL}(4, \mathbb{Z}/3\mathbb{Z})$.

(b) Show that there exists a nonabelian group of order $2025 = 3^4 \cdot 5^2$ in which the Sylow 3-subgroup is abelian.

5. Let K and H be groups, with K abelian. Consider a homomorphism $\phi : K \to \mathrm{Aut}(H)$ and the associated semidirect product $G = H \rtimes_\phi K$.

(a) Prove that when K is identified with the subgroup $\{(1, k) \mid k \in K\}$ of G, $\mathrm{Ker}(\phi)$ is contained in the center of G.

(b) Prove that every group of order 44 has a cyclic subgroup of order 22.

6. Let G be a finite group, and let P be a Sylow p-subgroup of G for some prime p. Let $N = N_G(P)$, the normalizer of P. Prove that $N_G(N) = N$.